How AI Sees

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: 0 3 255 blue

1. Color by number. Color every square with a **1**. Leave every **0** white.

0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	1	1	0	0
0	1	1	1	1	0	0	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	1	0	0
0	0	0	1	1	1	1	1	1	0	0	0
0	0	0	0	1	1	1	1	0	0	0	0
0	0	0	0	0	1	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0

What is the picture? _____

2. Pixels are numbers.

In a black-and-white picture, _____ is black and _____ is white.

In a color picture, each pixel is _____ numbers: red, green, and _____.

Word bank: corners edges neuron objects weights
You can use a word more than once.

3. A filter is a neuron.

A filter is a _____ with 9 inputs. Its 9 numbers are its _____.

0	0	255
0	0	255
0	0	255

$\times$

−1	0	+1
−1	0	+1
−1	0	+1

Total = _____ Does it fire? _____

If all 9 picture numbers are 255, the total is _____. Is there an edge? _____

4. Layers.

Layer 1 finds _____. Layer 2 looks at layer 1's maps, not the picture. It finds _____.

In the pictures from a real network, early layers find _____. Later layers find parts and whole _____.

Who chose the real network's filters? _____

5. Watch the layers. Count the corners yourself first. Then check the network.

Shape	My count	The network's count
square		
letter L		
letter T		
plus		
circle		

Why did the network find corners on the circle? _____

6. Talk about it. What could a self-driving car miss?

A self-driving car could make a mistake when _____

Go Further (Day 2). Find the edge by hand. Here is a tiny picture (0 = dark, 9 = bright) and a filter.

0	0	0	9	9	9
0	0	0	9	9	9
0	0	0	9	9	9

Filter:

−1	0	1
−1	0	1
−1	0	1

Put the filter over 3 columns of the picture. Multiply each pair of numbers, then add all 9.

Filter over columns	Math	Answer
1, 2, 3		
2, 3, 4		
3, 4, 5		
4, 5, 6		

Where are the big answers? _____